AP Precalculus
Unit 2A Day 10 Practice

Modeling with Sine and Cosine

For each word problem, first sketch a graph to represent the periodic motion. Then, fill out the appropriate characteristics and write a function to represent the graph.

1. As you ride a Ferris wheel, your distance from the ground varies sinusoidally with time. When the last seat is filled and the Ferris wheel starts, you find that it will take you 6 seconds to reach the bottom, 3 feet above the ground. The wheel makes one revolution every 18 seconds, and the diameter of the wheel is 44 feet.

   Period ______________ Sketch:

   Amplitude ______________

   Vertical Shift ______________

   Phase Shift ______________

   Equation:

2. Researchers found a creature from an alien planet. They noticed that its body temperature varied siusoidally with time. Thirty-five minutes after they start timing, it reaches a high of 120°F. Twenty minutes after that it reaches its next low, 104°F.

   Period ______________ Sketch:

   Amplitude ______________

   Vertical Shift ______________

   Phase Shift ______________

   Equation:

3. Mark Twain sat on the back of the Mississippi Queen and watched the paddlewheel. As the paddlewheel turned, he noticed that a point on the paddle blade moves in such a way that its distance, *d*, from the water's surface is a sinusoidal function of time. When his stopwatch read 5 seconds, the point was at its highest, 12 feet above the water's surface. The wheel's diameter was 14 feet and it completed a revolution every 16 seconds.

Period ____________ Sketch:

Amplitude ____________

Vertical Shift ____________

Phase Shift ____________

Equation:

4. As the motor on a jack turns, the walking beam rocks back and forth, pulling the pump rod in and out of a well. The pump is started at time $t = 0$ seconds. One second later, the rod is at its highest point 14 feet above the ground. It is at its next low point of 10 feet 2.5 seconds after that.

Period ____________ Sketch:

Amplitude ____________

Vertical Shift ____________

Phase Shift ____________

Equation: